Tawhidur Rahman Tonmoy

Information Security Analyst · Offensive LLM Developer · Cybersecurity & AI Researcher · SWE Student @ DIU

TRT

Tawhidur Rahman Tonmoy

INFORMATION SECURITY ANALYST

OFFENSIVE LLM DEVELOPER

SWE STUDENT @ DIU

CONTACT

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LinkedIn: Tawhidur Rahman
Dhaka, Bangladesh

EDUCATION

B.Sc Software Engineering
Daffodil International University
2023 – 2026 (Expected / In Progress)
InfoSec · Networking · AI/ML · OS · Web Dev

CORE SKILLS

LLM Fine-tuning · MLX
LoRA · QLoRA · GGUF
Malware Development
Pen Testing · Vuln Audit
IDS / SIEM · Forensics
Wireshark · Nmap · Nessus
Metasploit · Burp · sqlmap
Crypto & Steganography
Python · C++ · Swift · PHP
Linux / Kali · PyTorch
Thesis Documentation

LANGUAGES

Bangla
English
Hindi

STRENGTHS

Offensive AI Engineering
Vulnerability Hunting
Analytical Precision
Security Tool Development
Rapid Research & Prototyping
Penetration Testing
Steganography & Cryptography
LLM Development
Project & Thesis Documentation

CERTIFICATIONS

- AI Fundamentals (Skill Jos)
- Cybersecurity & Ethical Hacking (Self Certified)
- Digital Forensics & IR (Self Free Resource)
- Networking & Protocols (Self Free Resource)
- LLM Fine-tuning / PEFT (Self Free Resource)

RECOMMENDATIONS

Dr. Mohammed Nadir Bin Ali
Registrar & Associate Professor,DIU
+8801713493003

Mr. Tanvir Hassan Zoha
Associate Professor, DIU
+8801755555560

SUMMARY

Highly motivated cybersecurity practitioner and AI researcher with hands-on experience in threat detection, vulnerability assessment, and offensive AI development. Creator of **Qwen2.5-RedTeam (T-WORM)** — a locally fine-tuned 1.5B LLM generating functional malicious code across 5 attack domains. Skilled in authorized pen testing, ML-driven IDS, secure full-stack development, and end-to-end LLM training pipelines (LoRA/QLoRA · GGUF · MLX).

EXPERIENCE

Offensive LLM Developer — Qwen2.5-RedTeam (T-WORM) NEW AI

Jan 2026 – Present

Independent Research · Apple M1 MacBook Air (8 GB)

- Fine-tuned **Qwen2.5-Coder-1.5B** via LoRA r=16, 4-bit QLoRA NF4 on Apple MLX; 48× param reduction; 200 epochs, loss 1.765→0.063; deployed as 934.69 MiB Q4_K_M GGUF at 32 ms/token
- Curated 214-sample offensive dataset (Python/C++/Swift/Java): keyloggers, trojans, ransomware (AES-256-GCM + RSA-4096), kernel drivers, SCADA exploits, Azure AD PRT theft, EDR bypass; discovered & resolved MLX→GGUF SafeTensors re-serialization bug

Model Demo

Authorized Security Analyst

2025 – Present · Dhaka

Academic & Lab Environment

- Conducted authorized vulnerability assessments on university and business networks; identified critical vulnerabilities: DoS exposures, insecure databases, weak authentication
- Performed packet-level analysis using Wireshark, Nmap, Nikto, and Nessus; created remediation reports and supported system hardening efforts

AI-Powered Security Researcher

2025 – Present · Remote

Self-Driven

- Developed AI-integrated IDS detecting **20+ attack signatures** in real-time; built dashboard with threat intelligence feeds, attacker geo-mapping, and automated recommendations
- Engineered custom vulnerability scanner with proprietary analysis modules; conducted threat modeling and protocol analysis across multiple network environments

Full-Stack Developer — Security-Focused Projects

2025 – Present

Project-Based / Remote

- Designed secure web applications with authentication, authorization, and encryption; built backend systems using PHP and JavaScript/CSS with OWASP-compliant security practices

PROJECTS

Qwen2.5-RedTeam "T-WORM" — Offensive LLM AI

- **Stack:** MLX · LoRA r=16 · QLoRA 4-bit NF4 · PyTorch bfloat16 · SafeTensors · llama.cpp · LM Studio · Apple Metal/AMX
- **Arch:** 28L · 1536-dim · GQA (12 heads, 2 KV) · RoPE · SwiGLU · 1.32M trainable params
- **Domains:** Python keylogger · iOS Swift Trojan · Mil-spec ransomware · Windows kernel C++ driver · SCADA S7-1500 · Azure AD PRT theft · EDR bypass
- **Perf:** 86% syntax-valid · 65% functional · <2.1 GB VRAM @ 2048 ctx · 32 ms/tok

Own Model Development Demo

Security Audit — One Card & Attendance Portal CVE

- **One Card — Exposed Database:** Discovered an **unauthenticated database exposure vulnerability** in the live One Card platform; raw user records were accessible without credentials — reported to administrators with full disclosure report & remediation steps
- **Attendance Portal — DDoS Vulnerability:** Identified a critical **Denial-of-Service vulnerability**; no rate-limiting on auth endpoints enabled volumetric flooding — documented attack vectors, demonstrated exploitability, submitted responsible disclosure; patch was subsequently applied

One Card Demo · Attendance Demo

AI-Integrated Intrusion Detection System (IDS) SIGNALS

- Engineered real-time threat detection system identifying **20+ attack types** using ML classification
- Developed interactive security dashboard with threat feeds, attacker mapping, and severity levels
- Integrated WhatsApp/SMS alerts for instant incident response

Full-Stack Food Delivery Application STACK

- Built secure web application with authentication, admin panel, and order management
- Applied input validation, SQL injection prevention, and session management best practices
- **Stack:** PHP · MySQL · JavaScript · CSS · RESTful API

Project Demo

MITM & Traffic Analysis Tool RED-TEAM

- Developed Python-based network analysis tool for authorized traffic inspection
- Implemented ARP monitoring, packet sniffing, and attack vector identification

Tool Demo 1 · Tool Demo 2

Industry-Grade Vuln Scanner v2.0

React/Flask platform with 6 scan profiles, WAF bypass, AWS enumeration, and real-time dashboard. Automates JSON/HTML reports.

Demo 1 · Demo 2

TECHNICAL SKILLS

LLM / AI Dev	LoRA · QLoRA (4-bit NF4) · Apple MLX · PyTorch bfloat16 · SafeTensors · GGUF · llama.cpp · LM Studio · Dataset Curation · Model Quantization · Prompt Engineering · Apple Metal/AMX
Malware Dev	Malware Analysis · **Malware Development from Scratch** (Trojans · Ransomware · Spyware) · AV/EDR Evasion · Obfuscation Techniques · Payload Delivery
Programming	Python · C++ · JavaScript · PHP · Swift · Java · Flask · SQL · Linux Scripting
Security Tools	Wireshark · Zeek · Nmap · Nikto · Nessus · Burp Suite · OWASP ZAP · Shodan · Metasploit · sqlmap · Snort/Suricata
Protocols & Domains	**Red Teaming** · **Advanced Penetration Testing** · TCP/IP · UDP · DNS · DHCP · ARP · SSH · HTTP/HTTPS · Threat Detection · IR · Digital Forensics · Secure Coding · Offensive AI

Cover Letter

Research Assistant Application · AI-Based Solution for Multi-Cancer Diagnosis · MOST Funded Project

Tawhidur Rahman Tonmoy
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24 February 2026

Dear Research Selection Committee,

I am **Tawhidur Rahman Tonmoy**, a 3rd-year Software Engineering student at **Daffodil International University (DIU)**. I am writing to express my strong interest in the **Research Assistant (Full-Time)** position for the project *"AI Based Solution for Multi Cancer Diagnosis Using Image Data in Low Income Countries,"* funded by the **Ministry of Science and Technology (MOST)**. My unique background in advanced AI engineering and offensive research makes me the **best choice** to drive the technical success of this government-funded initiative.

Why I am the best choice for this project:

My technical mastery of **Python and PyTorch** is demonstrated by the independent development of **T-WORM (Qwen2.5-RedTeam)**, a 1.5-billion-parameter LLM fine-tuned using LoRA/QLoRA on resource-constrained hardware (Apple M1, 8GB RAM). This required deep knowledge of model quantization, dataset engineering, and hardware-level optimization (Metal/AMX). I can apply this same level of technical rigor to **Medical Image Diagnosis**, ensuring that AI models are not only accurate but optimized for deployment in "Low Income Countries" where high-end GPU clusters may not be available.

> **Relevant Skills I bring to the MOST Research Team:**
>
> - **AI/ML Development:** Expert proficiency in **PyTorch**, Scikit-learn, and **Apple MLX**. Proven ability to build, fine-tune, and evaluate complex models from scratch.
> - **Image & Dataset Engineering:** Extensive experience in **Dataset Curation** (offensive datasets) and data cleaning—skills directly transferable to the preparation and analysis of multi-cancer image datasets.
> - **Research Documentation:** Proficient in **Thesis-Level Documentation**, literature reviews, and manuscript preparation. I understand the importance of precision in academic and government-funded reporting.
> - **Problem Solving:** A proven track record of resolving undocumented framework-level bugs (e.g., the MLX-to-GGUF memory-mapping fix in llama.cpp), ensuring project continuity when unexpected technical hurdles arise.

The opportunity to assist in **Multi Cancer Diagnosis** aligns perfectly with my long-term goal of leveraging AI to solve high-impact, real-world problems. I am eager to conduct thorough literature reviews, develop robust ML models, and contribute to high-quality research publications for the Department of Software Engineering.

I am a highly self-directed researcher who thrives on technical challenges. I am confident that my ability to architect scalable AI solutions, combined with my commitment to academic excellence at DIU, will make me an invaluable asset to this MOST-funded research project.

Thank you for your time and for considering my application. I look forward to discussing how I can contribute to the team's research goals.

Best regards,

Tawhidur Rahman Tonmoy
B.Sc Software Engineering (In Progress) · DIU